

Big Multiplication

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Multiples of Ten

Get these automatic and the algorithm gets easy.

Warm-Up 11 ITEMS • RECALL

1 20×30

2 40×50

3 60×70

4 30×300

5 80×20

6 90×40

7 30×20

8 50×40

9 60×60

10 70×70

11 90×30

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. 400×60

13. 700×800

14. 25×40

15. 120×50

16. 36×200

17. 300×40

18. 500×600

19. 150×40

20. 220×30

21. 45×200

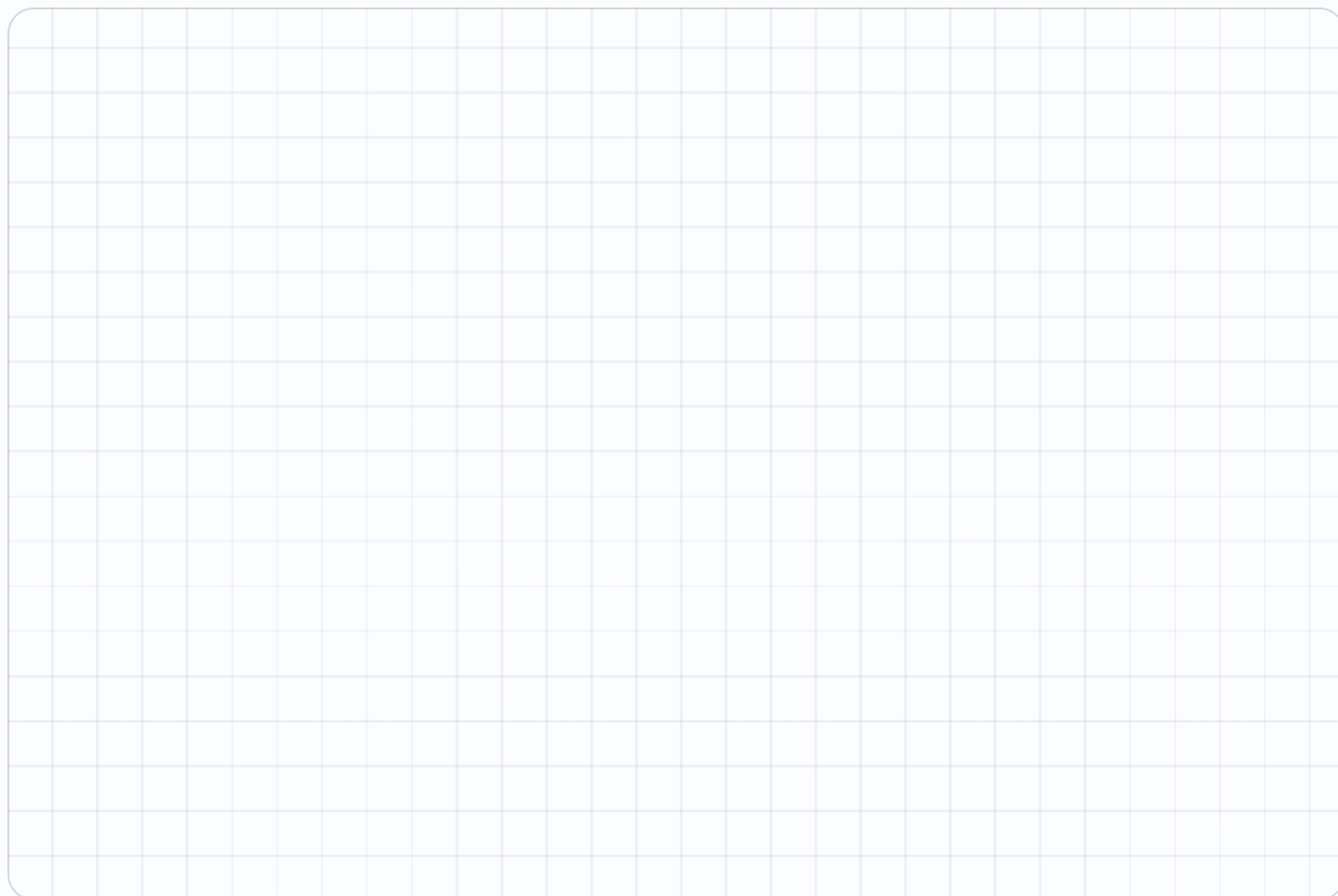
★ Challenge NOT GRADED • REASON IT OUT

22. 250×400
.....

23. 1250×80
.....

24. 1500×60
.....

25. 3500×200
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Partial Products

Four boxes, four numbers, one total.

Warm-Up 12 ITEMS • RECALL

1 23×10

2 23×4

3 Add 230 and 92

4 So 23×14 is

5 31×20

6 31×2

7 20×4

8 3×7

9 20×7

10 30×6

11 5×8

12 40×3

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 31×22

14. 45×23

15. 64×35

16. 128×12

17. 216×30

18. 24×6 as 20×6 plus 4×6

19. 32×7

20. 46×5

21. 58×4

22. 27×8

The Standard Algorithm

Stack it, but keep an estimate in your head.

✦ Warm-Up 12 ITEMS • RECALL

1 14×12

2 25×11

3 32×21

4 18×15

5 43×13

6 52×12

7 12×11

8 20×15

9 11×11

10 13×10

11 15×4

12 25×4

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 237×45

14. 418×27

15. 506×34

16. 725×18

17. 1234×12

18. 34×27

19. 52×19

20. 46×38

21. 63×24

22. 75×32

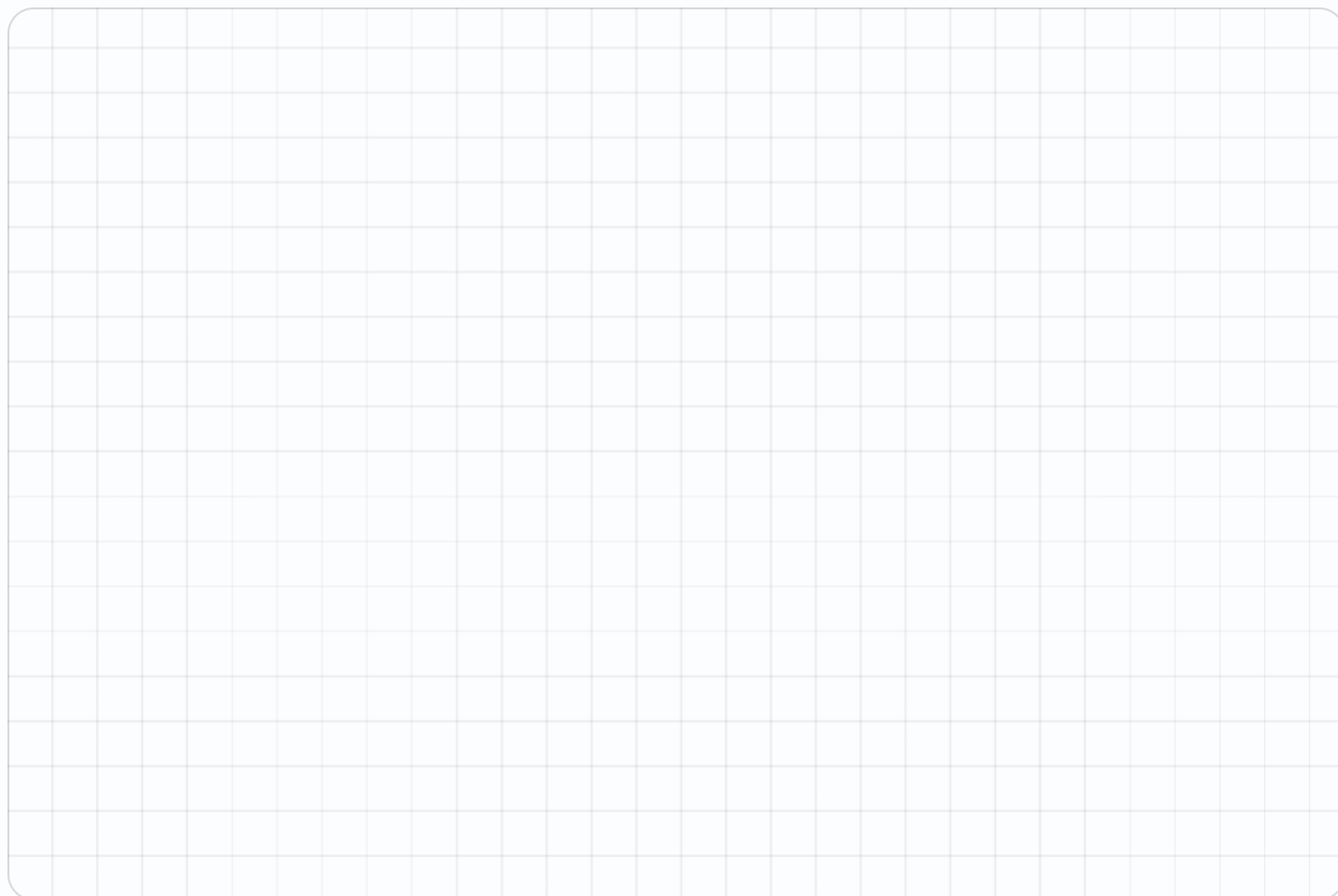
★ Challenge NOT GRADED • REASON IT OUT

23. 2450×36
.....

24. 3081×47
.....

25. 148×36
.....

26. 267×54
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Estimate to Check

Round both numbers, multiply, compare. A wrong answer should look wrong.

Warm-Up 12 ITEMS • RECALL

1 Estimate 19×21 by rounding both

2 Estimate 48×52

3 Estimate 31×29

4 Estimate 39×11

5 Estimate 62×18

6 Estimate 78×22

7 Estimate 21×19

8 Estimate 49×11

9 Estimate 32×28

10 Estimate 58×22

11 Estimate 81×19

12 Estimate 68×31

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19

.....

14. Estimate 289×31

.....

15. True value of 412×19

.....

16. Estimate $5,120 \times 48$

.....

17. Estimate 197×203

.....

18. Estimate 312×21

.....

19. True value of 312×21

.....

20. Estimate 489×29

.....

21. True value of 489×29

.....

22. Estimate 198×402

.....



Working space



Error journal

Poker & Floor Plan

Enrichment. Area two ways, and they must agree.

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦ Warm-Up

4 ITEMS • RECALL

1 Area of a 12 by 10 room, in squares

2 Area of a 20 by 15 room

3 40×50

4 12×12

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. A 14 by 12 room scaled by 3: new length

6. That scaled room's area (42 by 36)

7. 36×24

8. 143×22

★ Challenge NOT GRADED • REASON IT OUT

9. Digits 3, 4, 7, 2 — biggest product from two 2-digit numbers: type it (73 × 42)

10. Area of a 26 by 34 room

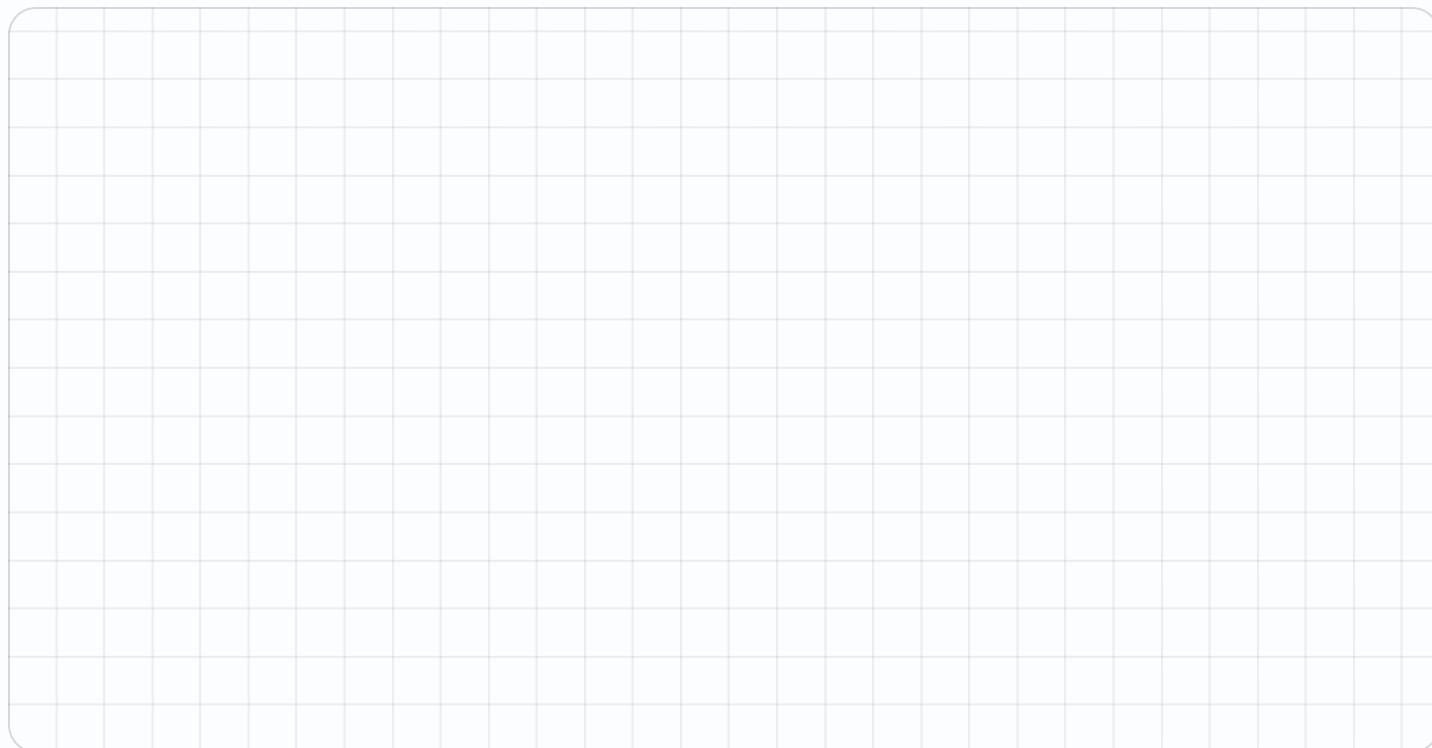
11. Double both sides of a 15 by 20 room. Area multiplies by

12. 1250×48

13. A 24 by 36 room — square feet

14. Doubling both sides of that room — the new area

Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal ONE SENTENCE PER MISTAKE

Mark the reasoning, not the handwriting.

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

1.1 • Multiples of Ten

OUT OF 21

Warm-Up • 1-11 • 600 | 2000 | 4200 | 9000 | 1600 | 3600 | 600 | 2000 | 3600 | 4900 | 2700

Core • 12-21 • 24000 | 560000 | 1000 | 6000 | 7200 | 12000 | 300000 | 6000 | 6600 | 9000

Challenge • 22-25 • 100000 | 100000 | 90000 | 700000

1.2 • Partial Products

OUT OF 22

Warm-Up • 1-12 • 230 | 92 | 322 | 322 | 620 | 62 | 80 | 21 | 140 | 180 | 40 | 120

Core • 13-22 • 682 | 1035 | 2240 | 1536 | 6480 | 144 | 224 | 230 | 232 | 216

Challenge • 23-26 • 7344 | 9022 | 744 | 1652

1.3 • The Standard Algorithm

OUT OF 22

Warm-Up • 1-12 • 168 | 275 | 672 | 270 | 559 | 624 | 132 | 300 | 121 | 130 | 60 | 100

Core • 13-22 • 10665 | 11286 | 17204 | 13050 | 14808 | 918 | 988 | 1748 | 1512 | 2400

Challenge • 23-26 • 88200 | 144807 | 5328 | 14418

1.4 • Estimate to Check

OUT OF 22

Warm-Up • 1-12 • 400 | 2500 | 900 | 400 | 1200 | 1600 | 400 | 500 | 900 | 1200 | 1600 | 2100

Core • 13-22 • 8000 | 9000 | 7828 | 250000 | 40000 | 6000 | 6552 | 15000 | 14181 | 80000

Challenge • 23-26 • 884 | 10000 | 200000 | 1472

Fri • Poker & Floor Plan

OUT OF 8

Warm-Up • 1-4 • 120 | 300 | 2000 | 144

Core • 5-8 • 42 | 1512 | 864 | 3146

Challenge • 9-14 • 3298 | 884 | 4 | 60000 | 864 | 3456

Multiples of Ten First

40 × 60 before 43 × 67. Every time.

Warm-Up 12 ITEMS • RECALL

1 20 × 30

2 40 × 60

3 50 × 50

4 70 × 30

5 80 × 40

6 90 × 90

7 30 × 30

8 20 × 50

9 60 × 40

10 70 × 80

11 90 × 20

12 50 × 60

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 400 × 70

14. 600 × 500

15. 250 × 40

16. 120 × 60

17. 35 × 200

18. 600 × 40

19. 700 × 300

20. 350 × 20

21. 140 × 50

22. 25 × 400

Estimate, Then Compute

Round both, multiply, then do it properly and compare.

Warm-Up 12 ITEMS • RECALL

1 Estimate 19×21

2 Estimate 48×52

3 Estimate 31×29

4 Estimate 62×18

5 Estimate 78×22

6 Estimate 39×11

7 Estimate 29×31

8 Estimate 41×62

9 Estimate 18×22

10 Estimate 77×33

11 Estimate 51×48

12 Estimate 91×12

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19

.....

14. True value of 412×19

.....

15. Estimate 289×31

.....

16. True value of 289×31

.....

17. Estimate 197×203

.....

18. Estimate 508×21

.....

19. True value of 508×21

.....

20. Estimate 392×41

.....

21. True value of 392×41

.....

22. Estimate 297×103

.....

★ Challenge NOT GRADED • REASON IT OUT

23. Estimate $5,120 \times 48$

24. Someone says $34 \times 26 = 68$. The real answer

25. Estimate $6,240 \times 38$

26. Someone says $52 \times 31 = 156$. The real answer

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Four Digits by Two

Longer, not harder. Keep the estimate in your head.

Warm-Up 12 ITEMS • RECALL

1 1000×10

2 2000×20

3 1200×10

4 3000×30

5 1500×20

6 2500×10

7 2000×10

8 1500×10

9 3000×20

10 4000×30

11 2500×20

12 1200×30

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 1234×12

14. 2450×36

15. 1875×24

16. 3081×47

17. 4206×15

18. 2143×13

19. 3260×24

20. 1450×32

21. 5012×26

22. 3105×18

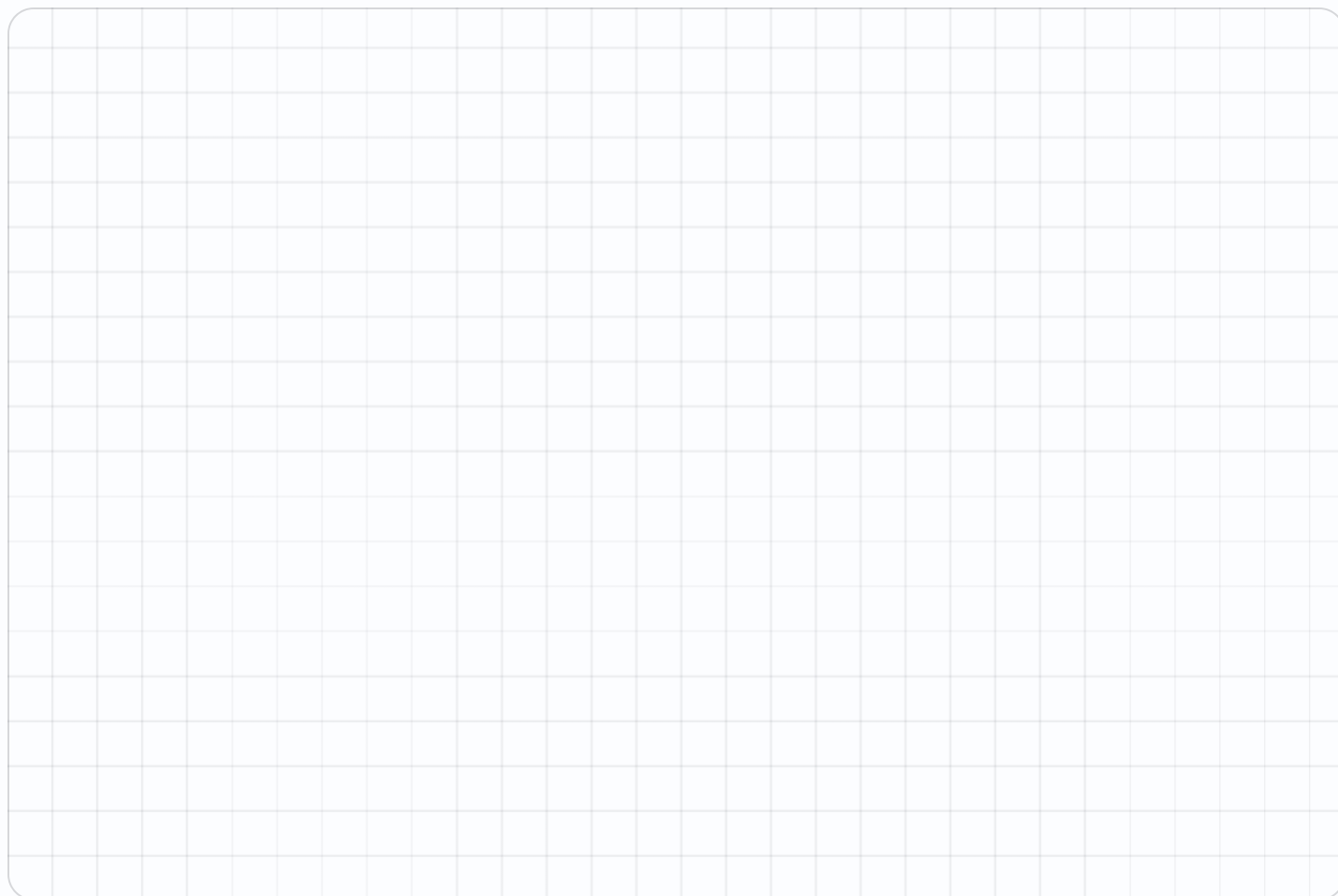
★ Challenge NOT GRADED • REASON IT OUT

23. 5678×43
.....

24. 9999×99
.....

25. 6789×34
.....

26. 8888×88
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Catch the Error

Given a wrong worked answer, find the line it went wrong.

Warm-Up 12 ITEMS • RECALL

1 23×14

2 23×4

3 23×10

4 $92 + 230$

5 31×22

6 45×23

7 34×12

8 34×2

9 34×10

10 $68 + 340$

11 41×21

12 52×13

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Someone got $23 \times 14 = 92$. What did they forget — type the missing partial

.....

14. Someone got $45 \times 23 = 135$. The real answer

.....

15. Someone got $38 \times 24 = 152$. The real answer

.....

16. Someone got $56 \times 27 = 392$. The real answer

.....

17. 63×48

.....

18. Someone got $34 \times 12 = 68$. The missing partial

.....

19. Someone got $52 \times 13 = 156$. The real answer

.....

20. Someone got $47 \times 26 = 282$. The real answer

.....

21. Someone got $63 \times 35 = 315$. The real answer

.....

22. 71×44

.....

Mid-Unit Quiz

Eight items across Weeks 1–2. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 40×60

2 23×14

3 50×40

4 34×12

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 45×23

6. 237×45

7. Estimate 412×19

8. 1234×12

9. 506×34

10. 52×13

11. 318×52

12. Estimate 508×21

13. 2143×13

14. 426×23

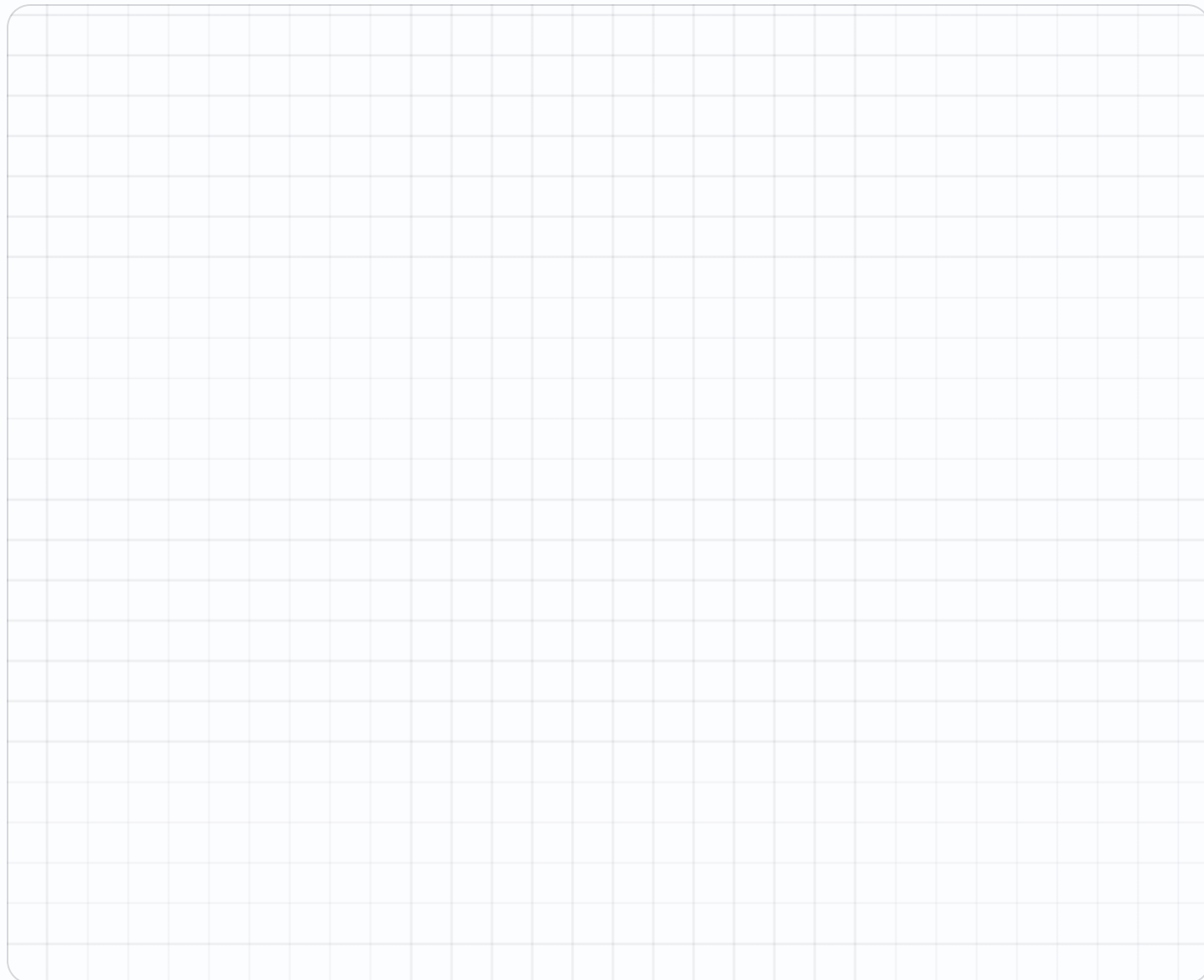
★ Challenge NOT GRADED • REASON IT OUT

15. 2450×36

.....

16. 3260×24

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 2 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

2.1 • Multiples of Ten First

OUT OF 22

Warm-Up • 1-12 • 600 | 2400 | 2500 | 2100 | 3200 | 8100 | 900 | 1000 | 2400 | 5600 | 1800 | 3000

Core • 13-22 • 28000 | 300000 | 10000 | 7200 | 7000 | 24000 | 210000 | 7000 | 7000 | 10000

Challenge • 23-26 • 100000 | 1000000 | 70000 | 900000

2.2 • Estimate, Then Compute

OUT OF 22

Warm-Up • 1-12 • 400 | 2500 | 900 | 1200 | 1600 | 400 | 900 | 2400 | 400 | 2400 | 2500 | 900

Core • 13-22 • 8000 | 7828 | 9000 | 8959 | 40000 | 10000 | 10668 | 16000 | 16072 | 30000

Challenge • 23-26 • 250000 | 884 | 240000 | 1612

2.3 • Four Digits by Two

OUT OF 22

Warm-Up • 1-12 • 10000 | 40000 | 12000 | 90000 | 30000 | 25000 | 20000 | 15000 | 60000 | 120000 | 50000 | 36000

Core • 13-22 • 14808 | 88200 | 45000 | 144807 | 63090 | 27859 | 78240 | 46400 | 130312 | 55890

Challenge • 23-26 • 244154 | 989901 | 230826 | 782144

2.4 • Catch the Error

OUT OF 22

Warm-Up • 1-12 • 322 | 92 | 230 | 322 | 682 | 1035 | 408 | 68 | 340 | 408 | 861 | 676

Core • 13-22 • 230 | 1035 | 912 | 1512 | 3024 | 340 | 676 | 1222 | 2205 | 3124

Challenge • 23-26 • 10665 | 9480 | 16536 | 15900

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 2400 | 322 | 2000 | 408

Core • 5-14 • 1035 | 10665 | 8000 | 14808 | 17204 | 676 | 16536 | 10000 | 27859 | 9798

Challenge • 15-16 • 88200 | 78240

Round to Estimate

Which way to round, and what it costs you.

Warm-Up 12 ITEMS • RECALL

1 Estimate 19×21

2 Estimate 48×52

3 Estimate 31×29

4 Estimate 62×18

5 Estimate 78×22

6 Estimate 39×11

7 Round 47 to the nearest ten

8 Round 312 to the nearest hundred

9 Round 89 to the nearest ten

10 Round 651 to the nearest hundred

11 Round 2,400 to the nearest thousand

12 Round 55 to the nearest ten

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19

.....

14. True value of 412×19

.....

15. Estimate 289×31

.....

16. True value of 289×31

.....

17. Estimate 197×203

.....

18. Estimate 47×31 by rounding both

.....

19. True value of 47×31

.....

20. Estimate 312×89

.....

21. True value of 312×89

.....

22. Estimate 651×48

.....



Working space



Error journal

Over or Under

Say before you compute whether your estimate is high or low.

Warm-Up 12 ITEMS • RECALL

1 Rounding both numbers up gives an estimate that is — over or under

2 Rounding both down gives — over or under

3 19×21

4 48×52

5 20×30

6 40×60

7 Rounding 47 up to 50 makes the estimate — type over or under

8 Rounding 42 down to 40 makes it — type over or under

9 Estimate 48×52 by rounding both to 50

10 True value of 48×52

11 Estimate 41×39

12 True value of 41×39

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 412×19 as 400×20

.....

14. True value

.....

15. Was the estimate over or under

.....

16. Estimate 78×22 as 80×20

.....

17. True value of 78×22

.....

18. Rounding both numbers up makes the estimate — type over or under

.....

19. Estimate 58×63 rounding both up

.....

20. True value of 58×63

.....

21. Estimate 29×31 rounding both to 30

.....

22. True value of 29×31

.....

Challenge

23. 63×48

24. Estimating that as $60 \times 50 = 3000$ — over or under

25. Why does rounding one up and one down land close — type cancel or add

26. Estimate 199×201 and give the true value



Working space



Error journal

Order of Magnitude

Is it hundreds, thousands or tens of thousands?

Warm-Up 12 ITEMS • RECALL

1 30×40

2 300×40

3 300×400

4 3×4

5 30×4

6 3000×4

7 Zeros in 100
_____8 Zeros in 1,000
_____9 20×30 — zeros in the answer

10 200×300

11 Zeros in 20×500

12 40×25

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Estimate 34×26 to the nearest hundred
.....14. Estimate 340×26
.....15. Estimate $3,400 \times 26$
.....16. Digits in the answer to 34×26
.....17. Digits in the answer to 340×26
.....18. About how big is 40×60 — type hundreds or thousands
.....19. About how big is 400×600 — type thousands or hundred thousands
.....20. Digits in the answer to 300×400
.....21. Digits in the answer to 50×50
.....22. Someone gets 812 for 40×60 . Digits it should have
.....



Mark Somebody's Work

Six worked answers, three wrong. Find them by estimating.

Warm-Up 11 ITEMS • RECALL

1 23×14

2 45×23

3 31×22

4 38×24

5 56×27

6 12×12

7 25×4

8 15×6

9 32×3

10 20×14

11 11×13

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. 23×14 given as 92 — the missing partial product
.....13. 45×23 given as 135 — the missing partial product
.....14. 237×45 given as 1,185 — the missing partial product
.....15. 56×27 with the second row as 112 — it should be
.....16. That answer is out by
.....17. Someone got $25 \times 14 = 100$. The real answer
.....18. Someone got $43 \times 21 = 129$. The real answer
.....19. Someone got $36 \times 25 = 180$. The real answer
.....20. Someone got $54 \times 32 = 1,068$. The real answer
.....21. 84×26
.....

Challenge NOT GRADED • REASON IT OUT

22. 78×46

23. 94×68

24. Someone got $209 \times 43 = 627$. The missing partial

25. 209×43

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Stadium Count Begins

Estimate from the photo before looking anything up.

Warm-Up 8 ITEMS • RECALL

1 24 seats × 38 rows _____	2 20 × 25 _____	3 500 × 16 _____	4 30 × 30 _____
5 40 × 25 _____	6 1000 × 16 _____	7 A section of 20 rows of 30 seats _____	8 Four such sections _____

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. 38 × 24	
10. 912 × 16	
11. Estimate 40 × 25 × 16	
12. The gap between that estimate and the true count	
13. 38 × 24 × 8	
14. A section of 42 rows of 38 seats	
15. Six such sections	

Challenge NOT GRADED • REASON IT OUT

16. $2,450 \times 36$

17. $1,875 \times 24$

18. A stadium of 24 sections of 1,850 seats

19. That stadium 80% full — seats taken

20. Estimate $24 \times 1,850$ by rounding

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Round to Estimate

OUT OF 22

Warm-Up • 1-12 • 400 | 2500 | 900 | 1200 | 1600 | 400 | 50 | 300 | 90 | 700 | 2000
| 60

Core • 13-22 • 8000 | 7828 | 9000 | 8959 | 40000 | 1500 | 1457 | 27000 | 27768 |
35000

Challenge • 23-26 • 250000 | 90000 | 160000 | 190548

3.2 • Over or Under

OUT OF 22

Warm-Up • 1-12 • over | under | 399 | 2496 | 600 | 2400 | over | under | 2500 | 2496
| 1600 | 1599

Core • 13-22 • 8000 | 7828 | over | 1600 | 1716 | over | 3600 | 3654 | 900 | 899

Challenge • 23-26 • 3024 | under | cancel | 39999

3.3 • Order of Magnitude

OUT OF 22

Warm-Up • 1-12 • 1200 | 12000 | 120000 | 12 | 120 | 12000 | 2 | 3 | 2 | 60000 | 4
| 1000

Core • 13-22 • 900 | 9000 | 90000 | 3 | 4 | thousands | hundred thousands | 6 | 4 | 4

Challenge • 23-26 • 884 | 10665 | 8 | 10000000

3.4 • Mark Somebody's Work

OUT OF 21

Warm-Up • 1-11 • 322 | 1035 | 682 | 912 | 1512 | 144 | 100 | 90 | 96 | 280 | 143

Core • 12-21 • 230 | 900 | 9480 | 1120 | 1008 | 350 | 903 | 900 | 1728 | 2184

Challenge • 22-25 • 3588 | 6392 | 8360 | 8987

Fri • Stadium Count Begins

OUT OF 15

Warm-Up • 1-8 • 912 | 500 | 8000 | 900 | 1000 | 16000 | 600 | 2400

Core • 9-15 • 912 | 14592 | 16000 | 1408 | 7296 | 1596 | 9576

Challenge • 16-20 • 88200 | 45000 | 44400 | 35520 | 40000

Count a Section

Seats per row, rows per section, sections per stadium.

Warm-Up 12 ITEMS • RECALL

1 24×2

2 24×10

3 24×20

4 24×30

5 24×38

6 12×24

7 15 rows of 20

8 25 rows of 20

9 18 rows of 10

10 30 rows of 30

11 12 rows of 25

12 40 rows of 15

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 26×42

14. 32×48

15. 28×36

16. 30×45

17. 22×54

18. 27 rows of 34

19. 46 rows of 28

20. 53 rows of 41

21. 38 rows of 45

22. 64 rows of 36

Challenge NOT GRADED • REASON IT OUT

23. $38 \times 24 \times 4$

24. $38 \times 24 \times 16$

25. A block of 125 rows of 48

26. That block minus 220 broken seats

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Scale It Up

One section times the number of sections.

Warm-Up 12 ITEMS • RECALL

1 900×10

2 900×16

3 1000×16

4 500×16

5 900×2

6 900×4

7 One section of 500, times 4

8 Times 10

9 600×8

10 250×4

11 1000×12

12 750×2

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 912×16

14. 912×8

15. $1,092 \times 12$

16. $1,536 \times 10$

17. $1,008 \times 20$

18. $1,850 \times 12$

19. $2,400 \times 18$

20. $3,150 \times 22$

21. $1,275 \times 16$

22. 960×45



Working space



Error journal

Explain the Gap

Compare to the published capacity and account for the difference.

Warm-Up 12 ITEMS • RECALL

1 16,000 – 14,592

2 900 – 884

3 15,000 – 14,592

4 16,000 – 15,000

5 1000 – 912

6 2500 – 2496

7 Estimated 2,000, actual 2,150 — the gap

8 Estimated 500, actual 480 — the gap

9 Estimated 1,000, actual 1,120 — the gap

10 6,000 – 5,850

11 Estimated 300, actual 300 — the gap

12 12,000 – 11,600

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Estimating 38×24 as 40×25 — over by

.....

14. That gap multiplied by 16

.....

15. 1,408 as a percent of 14,592, to the nearest whole

.....

16. Estimating 197×48 as 200×50 — the estimate

.....

17. True value of 197×48

.....

18. Estimated 40,000, actual 44,400 — the gap

.....

19. That gap as a percent of 44,400, to the nearest whole

.....

20. Estimated 27,000, actual 27,768 — the gap

.....

21. Rounding both numbers up gives an estimate that is — type high or low

.....

22. Estimated 9,000, actual 8,959 — the gap

.....



Challenge

NOT GRADED • REASON IT OUT

23. 197×203

24. An estimate of 40,000 is over by

25. A 4,400 gap on a 44,400 count — is the estimate useful, type yes or no

26. To halve that gap, round to the nearest — type hundred or thousand



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

✦ Warm-Up 10 ITEMS • RECALL

1 7×8

2 40×60

3 23×14

4 12×6

5 400×70

6 9×9

7 25×4

8 12×12

9 300×20

10 50×50

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 45×23

12. 237×45

13. 506×34

14. $1,234 \times 12$

15. Estimate 412×19

16. 34×27

17. 52×19

18. 1234×12

19. Estimate 489×29

Mission 02 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 50×50

2 38×24

3 30×40

4 21×12

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. 56×27

6. 289×31

7. 125×24

8. $2,450 \times 36$

9. Estimate $5,120 \times 48$

10. Estimate 197×203

11. $1,875 \times 24$

12. $9,999 \times 99$

13. 47×23

14. 312×21

15. Estimate 58×63

16. 2143×13

17. Someone got $52 \times 13 = 156$. The real answer



Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Count a Section

OUT OF 22

Warm-Up • 1-12 • 48 | 240 | 480 | 720 | 912 | 288 | 300 | 500 | 180 | 900 | 300 | 600

Core • 13-22 • 1092 | 1536 | 1008 | 1350 | 1188 | 918 | 1288 | 2173 | 1710 | 2304

Challenge • 23-26 • 3648 | 14592 | 6000 | 5780

4.2 • Scale It Up

OUT OF 22

Warm-Up • 1-12 • 9000 | 14400 | 16000 | 8000 | 1800 | 3600 | 2000 | 5000 | 4800 | 1000 | 12000 | 1500

Core • 13-22 • 14592 | 7296 | 13104 | 15360 | 20160 | 22200 | 43200 | 69300 | 20400 | 43200

Challenge • 23-26 • 15000 | 18240 | 116400 | 57800

4.3 • Explain the Gap

OUT OF 22

Warm-Up • 1-12 • 1408 | 16 | 408 | 1000 | 88 | 4 | 150 | 20 | 120 | 150 | 0 | 400

Core • 13-22 • 88 | 1408 | 10 | 10000 | 9456 | 4400 | 10 | 768 | high | 41

Challenge • 23-26 • 39991 | 9 | yes | hundred

Thu • Error Journal Sweep

OUT OF 19

Warm-Up • 1-10 • 56 | 2400 | 322 | 72 | 28000 | 81 | 100 | 144 | 6000 | 2500

Core • 11-19 • 1035 | 10665 | 17204 | 14808 | 8000 | 918 | 988 | 14808 | 15000

Challenge • 20-23 • 989901 | 88200 | 88200 | 244154

Fri • Mission 02 Test

OUT OF 20

Warm-Up • 1-4 • 2500 | 912 | 1200 | 252

Core • 5-20 • 1512 | 8959 | 3000 | 88200 | 250000 | 40000 | 45000 | 989901 | 1081 | 6552 | 3600 | 27859 | 676 | 22200 | 2304 | 9798

Challenge • 21-24 • 9480 | 14592 | 230826 | 57800